

Week 16 – Optimization of Compilation & Execution Pipelines

Summary:

Week 16 focused on **improving compilation and execution pipelines** for faster backend response.

Key Activities:

- Optimized **GCC/g++ compilation process** for C/C++ programs.
- Reduced **Python and Java execution overhead**.
- Tested **multi-threaded execution** for simultaneous program runs.
- Verified that Docker containers clean up properly after execution.

Technologies Used: Python, FastAPI, Docker, GCC, g++, JDK, WSL2

Outcome:

Backend is now **faster, more efficient**, and can handle multiple simultaneous executions reliably.

Flowchart:

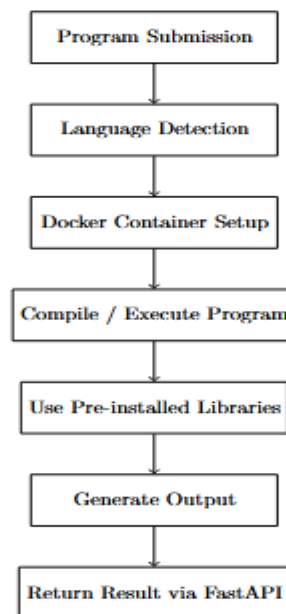


Figure 1: Simple Execution Flow of Multi-Language Compiler Backend